Huaze (Patrick) Liu

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Education

Harvey Mudd College 2022 – 2026
B.S. in Joint Computer Science & Mathematics (High Distinction) *Claremont, CA*

- **Cumulative GPA:** 3.732 / 4.00 (**Major GPA:** 3.915 / 4.00)
- **Relevant Coursework:** Data Structures, Algorithms, Computer Vision, Neural Networks, Machine Learning, Reinforcement Learning, Multivariable Calculus, Linear Algebra, ODEs, Intermediate Probability, Real Analysis, Optimization, Advanced Systems Engineering, Digital Signal Processing, State Estimation.

Publications

[C1] **Liu, H.***, Wu, Z.*, Jang, J.*, Sheng, J., Collins, A., Xie, A., Gu, Z., Zhu, K., Jiang, D., Cai, K., & Zhao, Y. (2026). *TAPNAV: Humanoid Navigation through Tactile Active Perception.* arXiv:2610.10748, in submission to ICRA 2027 (*equal contribution).

[C2] Fay, K.*, Shukla, A.*, **Liu, H.**, Djapri, D., Buynitsky, A., Hansen, N. (2026). *Catch It All!: Generalizable Dynamic Catching with a Dexterous Hand.* 4th RSS Workshop on Dexterous Manipulation, in submission to ICRA 2027 (*equal contribution).

[C3] **Liu, H.***, Gao, Z.*, Mohanty, A. (2025). *Language-Driven Semantic Change Detection in Urban Maps via Multi-Modal Deep Learning.* The 38th International Technical Meeting of the Satellite Division of the Institute of Navigation (ION GNSS+ 2025), Baltimore (*equal contribution).

[C4] **Liu, H.**, & Mohanty, A. (2025). *Conformal Prediction for Reliable Test-Time Uncertainty in Robotic Vision.* Southern California AI and Robotics Symposium (SCAIR 2025), Pomona.

Research Experience

Georgia Institute of Technology May 2026 – Present
Visiting Scholar, advised by Dr. Ye Zhao *Atlanta, GA*

- Developed a sim-to-real tactile active-perception planner and controller for vision-denied humanoid navigation. **[C1]**
- Developing an agentic humanoid system for long-horizon, contact-rich loco-manipulation that combines whole-body tactile sensing, task planning, and closed-loop control.

University of California, San Diego June 2025 – March 2026
Visiting Summer Scholar, advised by Drs. Hao Su and Michael T. Tolley *La Jolla, CA*

- Designed and trained a UniTree G1 upper-body interception policy under hardware torque and safety constraints. **[C2]**
- Implemented Sim-to-Real transfer using DAgger to deploy a vision-based controller at 50 Hz.

Harvey Mudd College July 2024 – Dec. 2025
Undergraduate Research Intern, advised by Dr. Adyasha Mohanty *Claremont, CA*

- Built a sensor-fusion map anomaly detection pipeline integrating image and LiDAR semantic segmentation. **[C3]**
- Evaluated split Conformal Prediction for segmentation and detection using vision foundation models. **[C4]**

Harvey Mudd College Aug. 2023 – Dec. 2025
Undergraduate Research Intern, advised by Dr. Calden Wloka *Claremont, CA*

- Built a video object-removal pipeline to automatically erase moving creatures from nature footage.
- Quantified the effects of padding-induced non-veridical representation on CNN performance with Mask R-CNN.

University of California, San Francisco May – Aug. 2023
UCSF ci2 Summer Fellow, advised by Dr. Srikantan Nagarajan *San Francisco, CA*

- Built Machine Learning models for Tinnitus classification using fMRI and MEG data, achieving 72.4% accuracy and benchmarking against Graph-CNN.

Work Experience

University of California, Berkeley Aug. 2025 – May 2026
Full-stack Machine Learning Engineer Intern *Berkeley, CA*

- Built a multi-pollutant Temporal Convolutional Network (TCN) pipeline in PyTorch for vehicle exhaust peak detection.
- Architected the data layer (TruckObject, DataWindow, checkpoint resolver) and an emission-factor calculator.
- Integrated the ML backend into a Flask web app with per-instrument model selection and cross-platform launchers.

Harvey Mudd College Jan. 2024 – Dec. 2025
Teaching Assistant *Claremont, CA*

- Tutored over 50 students in core concepts, provided homework assistance, guided projects, and graded assignments.
- Courses: CSCI153 – Computer Vision (2024 Fall, 2025 Spring, 2025 Fall), ENGR207 – Digital Signal Processing (2025 Spring), and CSCI070 – Data Structures & Program Development (2024 Spring, 2024 Fall).

Projects

ProxWeld for Vulcan OmniPro 220 | *TypeScript, Next.js* | GitHub Repo 2026
- Built a manual-grounded AI support agent for the Vulcan OmniPro 220 welder using retrieval over bundled PDF manuals and the Claude Agent SDK.
- Returned page-level citations, manual visuals, and structured wiring/troubleshooting helpers through a Next.js chat UI.

Automated Video Object Removal Tool | *Python* | GitHub Repo 2025
- Architected a full video object removal pipeline integrating tracking, SAM2 masks, and diffusion-based inpainting.
- Implemented temporal ghosting suppression post-process and improved video fidelity.

Vision-based Localization & Path Planning TurtleBot System | *Python, MATLAB* | Report 2025
- Developed a vision-based search TurtleBot 4 using EKF and UKF for state estimation and target tracking.
- Used the A* algorithm to compute and follow an optimal path to the GPS coordinates.

Multi-Objective Acquisition Function for Bayesian Optimization | *Python* 2025
- Developed a lightweight multi-objective acquisition function benchmarked on SVM and CNN, reducing latency by 8% compared to EI and UCB.

Vision Transformer for Traffic Object Detection | *Python* | GitHub Repo 2024
- Fine-tuned DETR on the KITTI dataset, achieving 71.0% mAP and 43.5% mAP (50:95) with 18.1 ms inference time.

Dynamic Time Warping with Confidence Intervals | *Python* | GitHub Repo 2024
- Applied Dynamic Time Warping to develop new metrics and visualizations for audio alignment confidence.

Spampede Game | *Java* | GitHub Repo 2023
- Designed a custom "Snake Game" UI and backend in Java using the MVC software design pattern; added an automated mode that finds optimal paths via BFS.

Skills

Programming Languages: Python, C++, SQL, C#, Java, MATLAB, LaTeX, R
Frameworks / Libraries: PyTorch, NumPy, Pandas, OpenCV, Scikit-learn, Hugging Face, Flask
Software / Tools: Git, Linux, ROS/ROS2, Isaac Sim/Lab, MuJoCo, SolidWorks, Docker, Jupyter, Visual Studio
Technical Skills: 2D/3D Computer Vision, Deep Learning, Sensor Fusion, Reinforcement Learning, Robotics Control
Hardware Experience: *Robots* – Franka, TurtleBot, Unitree G1, Booster T1, AgiBot G1; *Tactile Sensors* – TouchTronix, GelSight, Xela, Daimon

Awards & Grants

Harvey Mudd College Dean's List 2022 – 2026
Harvey S. Mudd Merit Award – Awarded $10,000 annually for academic merit ($40,000 total). 2022 – 2026
HMC Shanahan Projects Fund – Awarded $11,000 for an underwater autonomous pool-cleaning system. 2024 – 2025
HMC Leeds Student Travel Grant – Awarded $3,000 to attend the 38th ION GNSS+ Meeting, Baltimore, MD. 2025